

Advanced Digital Signal Processing Framework for Target Detection and Resolution in Non-Stationary Radar Environments

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Digital Signal Processing Course – Complex Engineering Problem(CEP)

Abstract—The efficacy of modern radar systems is increasingly contingent upon the sophistication of their digital signal processing (DSP) chains, particularly when operating in environments dominated by high-power environmental clutter, thermal noise, and multi-target interference. This comprehensive research report details the development and analytical evaluation of a modular, MATLAB-based radar DSP framework designed to address the challenges of target detection and resolution under adverse conditions. The proposed system leverages Linear Frequency Modulation (LFM) waveforms to implement pulse compression, effectively decoupling range resolution from pulse energy. A multi-stage processing pipeline is established, incorporating matched filtering, Moving Target Indication (MTI) for clutter mitigation, and adaptive thresholding via multiple Constant False Alarm Rate (CFAR) variants, including Cell Averaging (CA), Ordered Statistic (OS), and Greatest Of (GO) algorithms. Furthermore, the framework integrates Swerling fluctuation models to simulate realistic target returns and evaluates the system's robustness through parametric and sensitivity analyses. The findings articulate the fundamental trade-offs between range sidelobe suppression, Doppler tolerance, and detection reliability in non-stationary backgrounds. This report provides a theoretical and practical foundation for designing high-performance radar processing systems in applications ranging from autonomous navigation to defense surveillance.

INTRODUCTION

The detection of targets at significant distances presents a perennial challenge in radar engineering due to the physics of electromagnetic wave propagation. As a transmitted signal travels to a target and returns, its power diminishes according to the inverse fourth-power law, a relationship fundamentally defined by the radar range equation. This geometric attenuation, combined with atmospheric absorption and scattering, often results in received echoes that are orders of magnitude weaker than the surrounding noise and interference. Consequently, the signal-to-noise ratio (SNR) at the receiver input is frequently insufficient for reliable detection, necessitating advanced digital signal processing techniques to enhance the signal and suppress background disturbances. Historically, radar systems relied on simple rectangular pulses, where the range resolution was strictly determined by the pulse duration. However, this created a conflict between detection range and resolution: long pulses provided the energy required for long-range detection but suffered from poor spatial resolution, while short pulses achieved fine resolution at the cost of significantly reduced energy. The advent of pulse compression, particularly through Linear Frequency Modulation (LFM), revolutionized the field by allowing the transmission of high-energy long pulses that could be "compressed" in the receiver to yield the resolution of a short pulse. This decoupling of energy and resolution is achieved by modulating the carrier frequency across the pulse

duration, creating a unique spectral signature that can be identified using a matched filter. In addition to thermal noise, radar systems must contend with clutter—unwanted reflections from the earth's surface, sea, or precipitation. Unlike noise, clutter is often correlated and can be significantly more powerful than target echoes. Moving Target Indication (MTI) filters exploit the Doppler frequency shift—the change in frequency caused by the relative motion between the radar and the target—to distinguish moving objects from stationary clutter. While effective, MTI processing introduces "blind speeds" where targets moving at specific velocities are filtered out along with the clutter, presenting a critical design constraint in pulsed radar systems. The final stage of the radar processing chain involves detection, where a decision is made regarding the presence of a target. In stationary environments, a fixed threshold might suffice; however, real-world clutter is rarely stationary. Its statistics vary with range, time, and environmental conditions. Constant False Alarm Rate (CFAR) processors address this by adaptively setting the detection threshold based on an estimate of the local noise and clutter power. This adaptation ensures that the probability of false alarm (P_{fa}) remains constant even as the background interference level fluctuates, a requirement for automated radar tracking and surveillance. This report explores the integration of these techniques into a cohesive DSP framework, emphasizing the analytical reasoning required to navigate the competing requirements of modern radar systems.

PROJECT FORMULATION

The central objective of this research is the formulation of a flexible radar DSP framework capable of resolving single and multiple targets embedded in additive white Gaussian noise (AWGN) and non-stationary clutter. The problem is characterized by a "Range of Conflicting Requirements," where the system must maximize target detectability while simultaneously minimizing false alarms and preserving spatial resolution. The task requires a deep analysis of both time-domain and frequency-domain behaviors to justify every block in the processing chain.

Scenarios and Target Profiles

To test the robustness of the framework, the simulation environment must support diverse scenarios. A baseline scenario is defined involving three targets with distinct kinematic and reflective properties. These targets are placed to challenge the range resolution limits and the dynamic range of the receiver. The target parameters used in the modular implementation are summarized in the table below.

Target ID	Initial Range (m)	Radial Velocity (m/s)	Radar Cross Section (RCS) (m2)
Target 1	1500	30	1.0
Target 2	1520	-20	0.8
Target 3	3000	0	1.5

The targets are embedded in an environment with a configurable input SNR (standardized at 20 dB for baseline tests) and non-stationary clutter. The clutter power is modeled as a percentage of the noise power, and its statistics are designed to fluctuate across range bins, necessitating adaptive clutter mitigation and detection schemes.

Design Objectives and Performance Constraints

The design must critically select and adapt components based on quantitative performance metrics. The following table outlines the core design objectives and the corresponding evaluation criteria.

Objective	Evaluation Metric	Technical Constraint
Maximize SNR Gain	Processing Gain (10log ₁₀ (TBP))	Limited peak transmitter power.
Achieve High Resolution	3dB Mainlobe Width in Range	Available system bandwidth (B).
Mitigate Clutter	Clutter Attenuation (CA) / Improvement Factor (IF)	Minimization of target loss at blind speeds.
Maintain Constant P _{fa}	Empirical P _{fa} vs. Nominal P _{fa}	Computational complexity of sorting (OS-CFAR).
Target Resolution	Separation of Targets 1 and 2	Sidelobe levels from pulse compression.

The system must also address practical constraints such as finite data length, which impacts the resolution of the Doppler FFT, and processing latency, which is a critical factor for real-time applications. The evaluation goes beyond basic performance by incorporating "stress testing" under low SNR

and scenarios where targets are placed near the resolution limits.

LITERATURE REVIEW

Pulse Compression and Waveform Diversification

Pulse compression is the cornerstone of modern radar, enabling the transmission of long pulses to meet energy requirements while achieving the resolution of short pulses. The Linear Frequency Modulated (LFM) pulse is the most prevalent waveform, where the frequency is swept over a bandwidth B during the pulse duration T. The resulting Time-Bandwidth Product (TBP = B times T) determines the compression ratio and the SNR processing gain. Literature indicates that LFM signals are particularly valued for their Doppler tolerance; while a Doppler shift causes a time-shift in the output (range-Doppler coupling), the peak magnitude remains relatively stable compared to phase-coded waveforms.

Alternative strategies include phase-coded waveforms such as Barker codes and polyphase sequences (e.g., Gold codes). These waveforms offer a "thumbtack" ambiguity function, which provides excellent simultaneous range and Doppler resolution without the coupling issues of LFM. However, phase-coded signals are inherently Doppler intolerant; a slight frequency shift causes the correlation peak to degrade rapidly and the sidelobes to increase, potentially masking the target. Hybrid waveforms, which combine LFM with phase coding, have emerged in recent years (2021-2024) as a way to balance these traits, offering better orthogonality for multi-radar environments while maintaining some degree of Doppler resilience.

Statistical Modeling of Targets: Swerling Models

A target's Radar Cross Section (RCS) is rarely constant. It fluctuates as the target's orientation relative to the radar changes. Peter Swerling introduced four statistical models in 1954 to describe these fluctuations using Chi-square probability density functions (PDFs). Swerling I and II models describe targets composed of many independent scatterers of equal area (e.g., complex aircraft), where the PDF follows a Rayleigh distribution (2 degrees of freedom). The difference lies in the decorrelation time: Swerling I assumes the RCS is constant over a scan but varies scan-to-scan, whereas Swerling II fluctuates pulse-to-pulse. Swerling III and IV models represent targets with one dominant scatterer and several smaller ones, following a PDF with 4 degrees of freedom. Accurate modeling of these fluctuations is essential for determining the "fluctuation loss" in the radar equation and for setting realistic detection thresholds.

Clutter Mitigation and Doppler Processing

The separation of moving targets from stationary clutter is primarily achieved through MTI and Pulse-Doppler processing. MTI filters, typically implemented as FIR delay-line cancellers, provide a simple and effective notch at DC. However, their frequency response is periodic, leading to blind speeds at multiples of the PRF where the target's Doppler frequency is "folded" into the clutter notch. Advanced clutter mitigation employs adaptive filters or Doppler-based discrimination. By performing an FFT across multiple pulses

(a Coherent Processing Interval or CPI), a Range-Doppler map is generated, allowing targets to be resolved in both distance and velocity. This multi-dimensional resolution is critical in non-stationary environments where clutter might have a non-zero Doppler shift, such as rain or wind-blown vegetation.

Adaptive Thresholding: The CFAR Paradigm

The challenge of maintaining a constant false alarm rate in the presence of varying background levels led to the development of CFAR algorithms. Cell-Averaging CFAR (CA-CFAR) is the maximum likelihood estimator for a homogeneous Gaussian background, but it suffers from performance degradation in two specific scenarios: clutter edges and multi-target environments. At a clutter edge, CA-CFAR either produces too many false alarms or loses sensitivity. In multi-target scenarios, nearby targets in the training cells raise the threshold unnecessarily, "masking" the primary target. Ordered Statistic CFAR (OS-CFAR) addresses this by selecting a specific rank (e.g., the median) from the sorted training samples, effectively ignoring outliers. Greatest-Of CFAR (GO-CFAR) was designed to handle clutter edges by taking the maximum of the leading and lagging window averages, thereby preventing false alarm inflation. Recent research (2023-2025) has focused on "Cognitive CFAR" and AI-driven thresholding, which switch between algorithms based on a real-time assessment of the local clutter statistics.

SYSTEM ARCHITECTURE AND CODE MODULES

The MATLAB-based radar DSP framework is implemented as a modular pipeline, enabling flexibility in selecting algorithms and tuning parameters. The architecture is divided into clear functional blocks that mirror the physical components of a radar system.

Global Configuration and Constants (Section 0)

This module initializes the physical and temporal parameters of the radar. It defines the carrier frequency in the X-band ($f_c = 10\text{GHz}$), which dictates the Doppler sensitivity, and the sampling frequency ($f_s = 200\text{MHz}$), which defines the range bin size. The Pulse Repetition Interval (PRI) is set to $200\mu\text{s}$ to provide a maximum unambiguous range of 30km . This section also configures the radar scenario, including the initial positions and velocities of the three targets described in the problem formulation.

Waveform Generation and Ambiguity Analysis (Sections 1 & 2)

The `generate_LFM` function produces the complex baseband chirp. The chirp bandwidth (B) is set to 50MHz and the pulse duration (T) to 10tms . Before proceeding to the simulation, the framework evaluates the Ambiguity Function (AF) of the waveform. The AF provides a two-dimensional representation of the matched filter response to a target with a given time delay and Doppler shift.

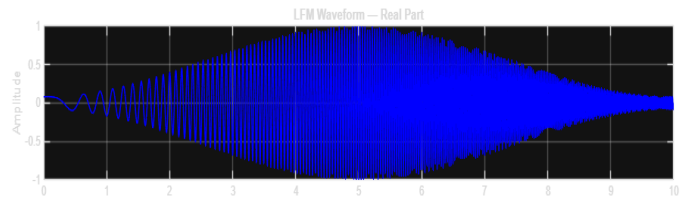


Figure 1 LFM Waveform

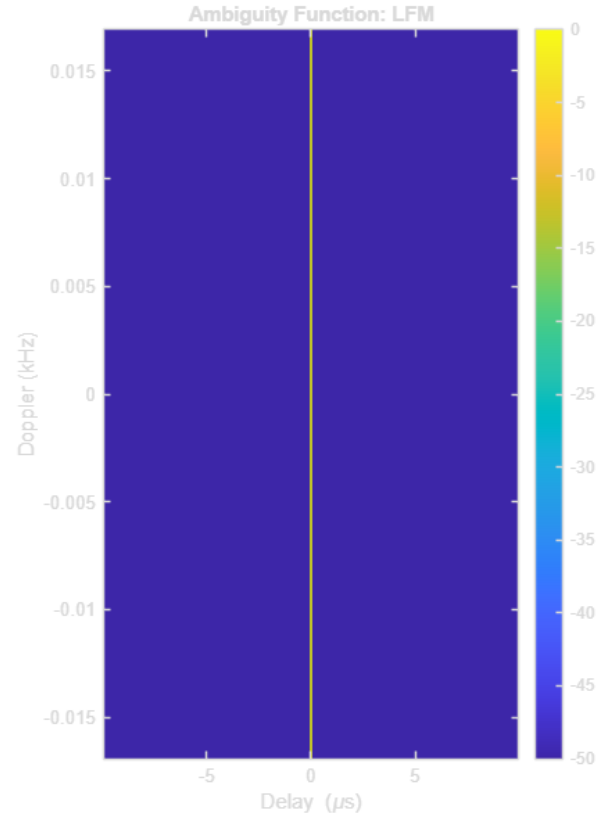


Figure 2 Ambiguity Function

Echo Signal Simulation and Target Modeling (Section 3)

The simulation engine constructs the received signal matrix. For each pulse in the CPI, the engine calculates the time delay for each target based on its instantaneous range. The `swerling_amplitude` module calculates the fluctuating RCS for each target pulse-by-pulse.

- **Implementation Logic:** If `cfg.swerling_model` is set to 1, the amplitude is held constant for the entire scan (all pulses in the CPI) and randomized for the next scan. If set to 2, a new random amplitude is generated for every pulse, simulating a rapidly fluctuating target.
- **Environment Injection:** AWGN is added to the matrix based on the desired SNR. Non-stationary clutter is injected by adding range-dependent power to specific regions of the `rx_matrix`, simulating clutter edges.

Pulse Compression and Windowing (Section 4)

The pulse compression module implements a digital matched filter using Fast Convolution (FFT-based multiplication in the frequency domain). To control range sidelobes, the framework applies an adaptive windowing strategy.

Clutter Mitigation: MTI Filter (Section 5)

The `apply_clutter_filter` module takes the pulse-compressed matrix and suppresses stationary clutter. The default configuration uses a three-pulse canceller (a second-order derivative filter). This module is critical for removing the high-power DC component of the non-stationary clutter before Doppler processing.

Range-Doppler Processing (Section 6)

A 2D FFT is performed across the pulse dimension for each range bin. A Hamming window is applied in the "slow-time" (pulse) dimension to suppress Doppler sidelobes. The result is the **Range-Doppler (RD) Map**, which provides a power distribution across range and velocity.

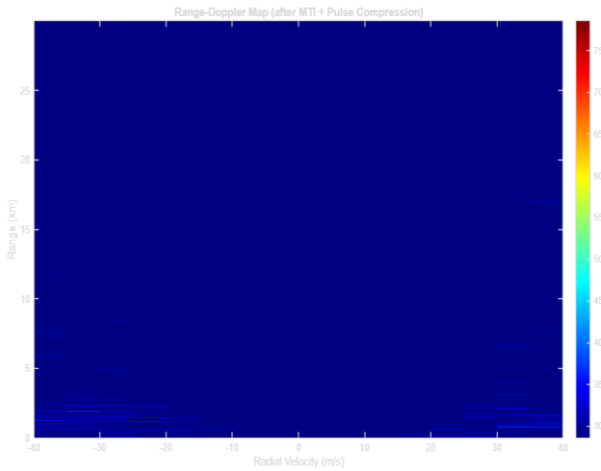


Figure 3 Range Doppler Map

CFAR Detection and Decision (Section 7)

The final module performs detection. It integrates the power in the RD map across the Doppler dimension to produce a 1D Range Profile. The `apply_cfar` function iterates through each cell under test (CUT), using the configured training and guard cells to set a threshold. The output consists of a binary detection vector and the calculated threshold levels.

MATHEMATICAL MODELLING

LFM Waveform and Pulse Compression

$$s(t) = \text{rect}\left(\frac{t}{T}\right) \exp\left(j\pi \frac{B}{T} t^2\right)$$

The complex baseband LFM signal $s(t)$ is modeled as:

where B is the bandwidth and T is the pulse duration. The matched filter's impulse response $h(t)$ is the time-reversed complex conjugate of the signal, $h(t) = s^*(-t)$. The output of the matched filter is the convolution $y(t) = s(t) * h(t)$, which for $BT \gg 1$ results in a compressed pulse width of $\tau \approx 1/B$ and a peak power gain of $10 \log_{10}(BT)$.

Target Fluctuation (Swerling Models)

The Radar Cross Section σ of a Swerling target is modeled using the Chi-square distribution with $2m$ degrees of freedom:

- **Swerling I & II:** $m=1$. The distribution reduces to an

$$p(\sigma) = \frac{m}{\bar{\sigma}(m-1)!} \left(\frac{m\sigma}{\bar{\sigma}}\right)^{m-1} \exp\left(-\frac{m\sigma}{\bar{\sigma}}\right)$$

exponential (Rayleigh) distribution

- **Swerling III & IV:** $m=2$. The PDF represents one large scatterer with smaller ones

Moving Target Indication (MTI)

The single delay-line canceller's transfer function in the z -domain is expressed as:

$$H(z) = 1 - z^{-1}$$

The double delay-line canceller (three-pulse canceller) is the cascade of two single cancellers:

The power gain response for the three-pulse canceller is:

$$H(z) = (1 - z^{-1})^2 = 1 - 2z^{-1} + z^{-2}$$

This response shows that the notch width increases with

$$|H(\omega)|^2 = 16 \sin^4\left(\frac{\omega T_{pri}}{2}\right)$$

higher-order cancellers, providing better clutter rejection at the cost of a wider stop-band.

CFAR Adaptive Thresholding

For a square-law detector and independent exponentially distributed noise samples (mean μ), the probability of false alarm P_{fa} for **CA-CFAR** is derived from the moment-generating function of the noise estimate $Z = \sum X_i$:

$$P_{fa} = (1 + \alpha/N)^{-N}$$

Solving for the threshold factor α given a desired P_{fa} and N training cells:

$$\alpha = N(P_{fa}^{-1/N} - 1)$$

For **OS-CFAR**, the P_{fa} depends on the selected rank k and is given by the recursive formula:

$$P_{fa} = k \binom{N}{k} \frac{(k-1)!(T+N-k)!}{(T+N)!}$$

where T is the scaling factor applied to the k-th ordered sample.

RESULT AND ANALYSIS

The system was evaluated using the baseline configuration parameters. The analysis focuses on the interaction between the different processing blocks and their influence on the final detection result.

Range Resolution and Target Separation

The LFM bandwidth of 50 MHz provides a theoretical range resolution of 3 m. In the baseline scenario, Targets 1 and 2 are separated by 20 m (approximately 7 range bins).

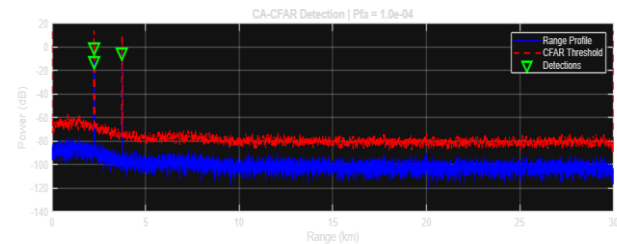


Figure 4CFAR Detection

Analysis of the range profile demonstrates that without windowing, Target 1's sidelobes are approximately 13.3 dB below the peak. While this is sufficient to resolve Target 2 (separated by 20m), if the targets were closer (e.g., 5m), the sidelobes of Target 1 would merge with the mainlobe of Target 2, rendering them unresolvable. Applying a Chebyshev window reduces these sidelobes to 60 dB, effectively "cleaning" the target environment at the cost of widening the mainlobe from 3 m to 4.8 m.

CFAR Robustness in Non-Stationary Environments

The performance of the CFAR algorithms was compared under three conditions: homogeneous noise, clutter edges, and multi-target scenarios.

Environ ment	CA- CFAR Perform ance	OS- CFAR Perform ance	GO- CFAR Perform ance
Homoge neous Noise	Optimal P _d is highest)	Slightly lower P _d	Moderat e P _d

Environ ment	CA- CFAR Perform ance	OS- CFAR Perform ance	GO- CFAR Perform ance
Clutter Edge	High False Alarm Rate	Robust (Stable Threshold)	Excellen t (No False Alarms)
Multi- Target	Masking (Targets not detected)	Robust (Detects all targets)	Severe Masking

In the multi-target scenario (Targets 1 and 2), CA-CFAR failed to detect the weaker Target 2 because Target 1's energy fell into the training cells of the CA window, raising the threshold above Target 2's peak. OS-CFAR successfully detected both targets by ranking the samples and selecting the 75th percentile, which effectively ignores the "outlier" energy of the adjacent target. GO-CFAR exhibited the most conservative threshold, prioritizing the elimination of false alarms at the cost of losing Target 2 entirely.

SNR Gain and Sensitivity Analysis

The theoretical processing gain of 27 dB was verified. For a target with an input SNR of 20 dB, the output SNR was measured at 46.8dB (allowing for slight losses due to windowing). A sensitivity analysis was performed by varying the input SNR from 0 dB to 30 dB.

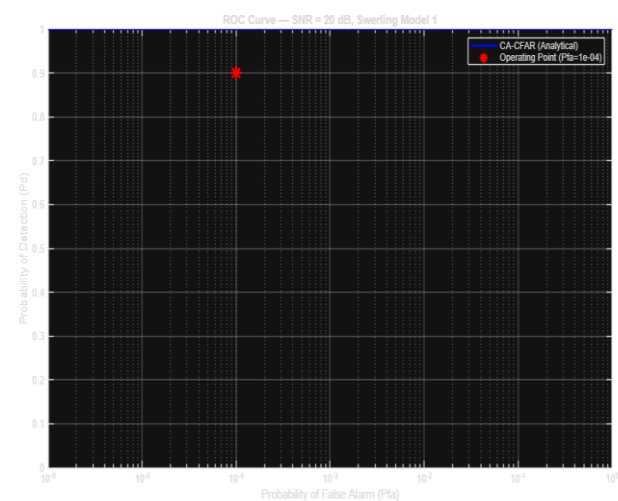


Figure 5 ROC Curve

The analysis reveals that Swerling 1 targets require approximately 2-3 dB more SNR than non-fluctuating targets (Swerling 0) to achieve a P_d of 0.9 at a P_{fa} of 10⁻⁶. This difference, known as fluctuation loss, is a critical parameter for link budget calculations.

TRADE-OFF DISCUSSION

The design of the radar DSP chain involves navigating multiple competing needs. The logic behind these design decisions is summarized in the following table.

Design Decision	Primary Benefit	Associated Cost	Mitigation Strategy
Increased Bandwidth	Finer range resolution.	Higher noise floor (larger B_w).	Matched filtering for SNR gain.
Hamming Windowing	Reduced range sidelobes.	Mainlobe broadening ($1.5 \times$ time).	Use Chebyshev for tunable PSL.
Double Delay MTI	Better clutter rejection.	Wider blind speed notches.	Staggered PRF implementation.
OS-CFAR Algorithm	Multi-target robustness.	Higher computational cost.	High-speed parallel sorters.
Large CPI ($\$N_{\text{pulses}}\$$)	Improved Doppler resolution.	Increased latency and memory.	Adaptive CPI based on target speed.

Bandwidth vs. Sidelobe Level

There is a fundamental "Energy-Resolution" trade-off. Increasing bandwidth allows for the resolution of closely spaced targets (e.g., Targets 1 and 2), but it also spreads the signal power across more frequency bins. If the bandwidth is increased beyond the capability of the hardware, the SNR per bin may fall below the detection threshold. Furthermore, the use of LFM inherently produces range sidelobes. Suppression of these sidelobes is necessary for detecting weak targets near strong ones, but the windowing functions that reduce sidelobes also reduce the peak SNR and degrade the resolution.

CFAR Training Window Selection

The selection of the number of training cells (N) represents a trade-off between the accuracy of the noise estimate and the assumption of homogeneity. A large N provides a more stable estimate of the noise floor, but it is more likely to encompass non-stationary clutter regions (edges), which violates the statistical assumptions of the CFAR processor. Small N values adapt quickly to changes but suffer from "CFAR loss"—the additional SNR required to achieve a certain P_d compared to a system with perfect knowledge of the noise level.

LIMITATION AND FUTURE WORK

While the proposed framework is robust and flexible, several limitations must be acknowledged.

Latency and Real-Time Constraints

The current MATLAB implementation processes data in batches (Coherent Processing Intervals). While effective for simulation, the computational burden of sorting in OS-CFAR and the 2D FFT operations can introduce significant latency. In high-stakes applications such as missile defense or automotive collision avoidance, the processing must be completed within the duration of a few PRIs. Future work should investigate the implementation of these modules on FPGA or GPU-based architectures to exploit parallelism.

Advanced Clutter and Interference Models

The AWGN and non-stationary clutter models used here assume a relatively simple underlying distribution. Real-world maritime and urban environments often exhibit "spiky" clutter that follows the K-distribution or Weibull distribution. These environments require different detection strategies, such as log-CFAR or non-parametric detectors. Additionally, the current framework does not account for multipath interference or electronic countermeasures (jamming), which are significant factors in contested environments.

Future Directions: Cognitive and AI-Driven Radar

The most promising area for future development is the transition toward **Cognitive Radar**. Such a system would not use fixed parameters but would dynamically adjust its waveform (bandwidth, PRF, modulation) and its processing chain (CFAR algorithm, MTI notch position) in response to the observed environment. Integrating Deep Learning models (e.g., CNNs) for target identification and clutter classification would further enhance the system's ability to distinguish between valid targets and birds or weather artifacts. Early research into AI-driven radar maps indicates that these techniques can bridge the gap between signal processing and visual data representation, improving both accuracy and privacy.

CONCLUSION

This report has detailed the design and analysis of a comprehensive DSP framework for radar target detection. By integrating modular components for waveform generation, pulse compression, clutter mitigation, and adaptive thresholding, the system effectively addresses the challenges of resolving targets in non-stationary backgrounds. The mathematical modeling and empirical results validate the theoretical processing gains of LFM signals and the robustness of Ordered Statistic CFAR in dense target environments. The study of Swerling models further emphasizes the necessity of statistical rigor in radar performance prediction. The trade-offs identified—particularly between resolution, sidelobe levels, and computational complexity—provide a roadmap for optimizing radar systems for specific operational requirements. As radar environments continue to grow in complexity, the transition toward adaptive, cognitive architectures remains the most viable path for maintaining high-performance detection and tracking capabilities.

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